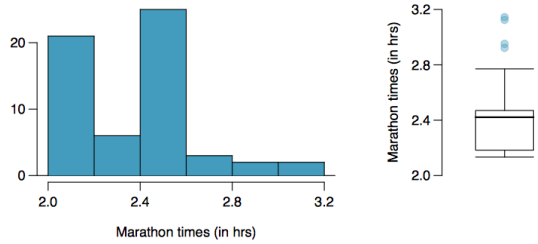
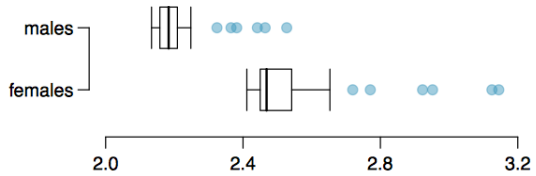
STA220 Fall 2015 Additional Problems - Week 1 Modules

1. (Source: *OpenIntro Stats* 1.27) In a class of 25 students, 24 of them took an exam in class and 1 student took a make-up exam the following day. The professor graded the first batch of 24 exams and found an average score of 74 points with a standard deviation of 8.9 points. The student who took the make-up the following day scored 64 points on the exam.
   (a) Does the new student's score increase or decrease the average score?
   (b) What is the new average?
   (c) Does the new student's score increase or decrease the standard deviation of the scores?

2. (Data from *OpenIntro Stats* 1.40) The histogram and box plots below show the distribution of finishing times for male and female winners of the New York Marathon between 1980 and 1999.



   (a) What features of the distribution are apparent in the histogram and not the box plot? What features are apparent in the box plot but not in this histogram?
   (b) How will the distribution of the z-scores of winning times compare?
   (c) Consider the distribution of marathon times for men and women based on the box plots below.



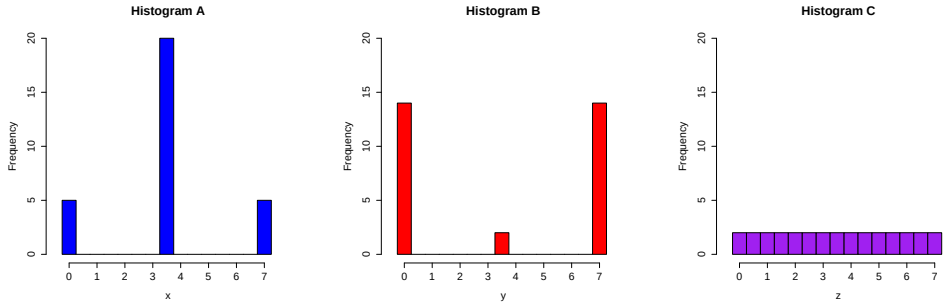
      i. For which sex is the distribution most skewed?
      ii. Which is larger, the maximum winning time for males or the first quartile winning time for females?
      iii. For which sex is the variability largest?
      iv. For females, would the mean be larger or smaller than the median?

3. The Salk polio vaccine trial was carried out in 1954. The table below shows the incidence of polio for 400,000 participating children, half of whom received the vaccine and half who did not.

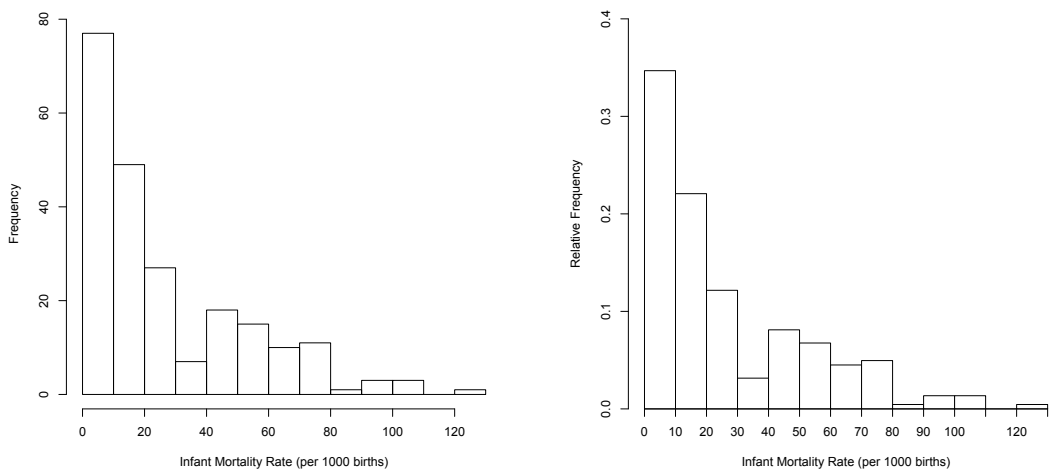
	Polio victim?	
	Yes	No
Placebo	142	199,858
Salk Vaccine	56	199,944

   (a) What percent of patients got polio and what percent did not? What type of distribution do these numbers comprise?
   (b) What percent of patients in the vaccine group got polio? What percent of patients in the placebo group got polio? From what distributions are these percentages?
   (c) Is there a relationship between whether or not a child was a polio victim and whether or not they got the vaccine? What distribution supports your answer?

4. Order the data displayed in the three histograms from least to most variable.

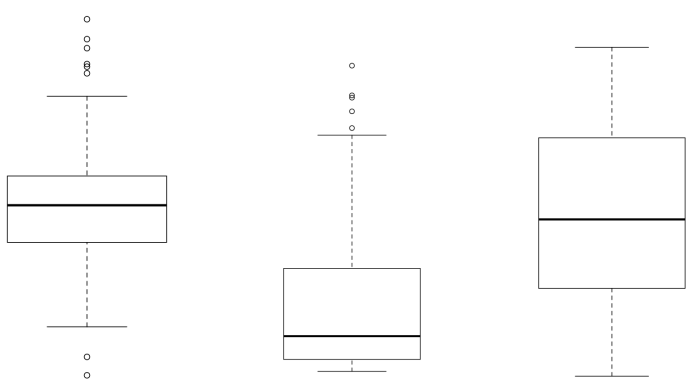


5. (Data from *OpenIntro Stats* 1.36)
The infant mortality rate is defined as the number of infant deaths per 1,000 live births. The data in the histograms below are the estimated infant mortality rates in 2012 for 222 countries. Note that one histogram displays frequencies, and the other displays relative frequencies.



(a) Which of these words are useful to describe the shape of the distribution: symmetric, right-skewed, left-Skewed, bimodal, uniform, large outliers?

(b) Estimate the following: minimum, Q1, median, Q3, maximum

(c) Do you expect the mean to be larger or smaller than the median? Why?

(d) Which boxplot corresponds to the infant mortality data?